EBN 111/122
ELECTRICITY AND ELECTRONICS

Chapter 10
Sinusoidal Steady-State Analysis

1

Sinusoidal Steady-State Analysis Chapter 10

2

10.1 Basic Approach (1)

Steps to Analyze AC Circuits:

1. Transform the circuit to the phasor or frequency domain.
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, superposition, etc.).
3. Transform the resulting phasor to the time domain.

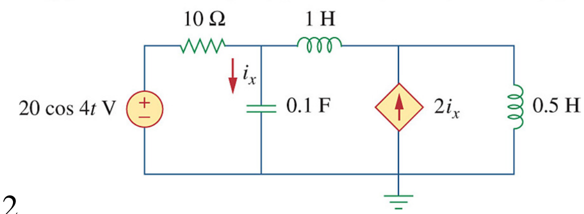
Time to **Phasor** → Solve variables in **Phasor** → **Phasor** to Time

3

10.2 Nodal Analysis

Example 10.1

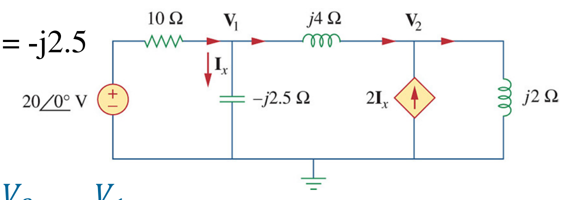
Determine i_x.



$jX_L = j\omega L = j \times 4 \times 1 = j4$

$jX_L = j\omega L = j \times 4 \times 0.5 = j2$

$jX_C = -j/\omega C = -j \div (4 \times 0.1) = -j2.5$



KCL at node 1:

$$\frac{V_1 - 20\angle 0^\circ}{10} + \frac{V_1 - V_2}{j4} + \frac{V_1}{-j2.5} = 0$$

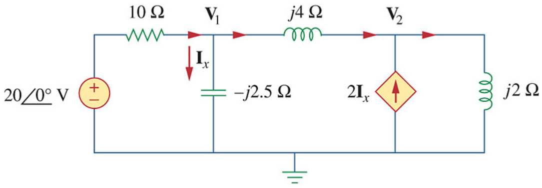
(x20) $2V_1 - 40 - j5(V_1 - V_2) + j8V_1 = 0$

$$(2 + j3)V_1 + j5V_2 = 40 \quad 1$$

4

Example 10.1 cont

Determine i_x.



KCL at node 2:

$$\frac{V_2 - V_1}{j4} - 2Ix + \frac{V_2}{j2} = 0$$

But: $I_x = \dfrac{V_1}{-j2.5}$ hence: $\dfrac{V_2 - V_1}{j4} - 2\left(\dfrac{V_1}{-j2.5}\right) + \dfrac{V_2}{j2} = 0$

$$(\times j20)\ \ 5V_2 - 5V_1 + 16V_1 + 10V_2 = 0$$

$$11V_1 + 15V_2 = 0 \quad (2)$$

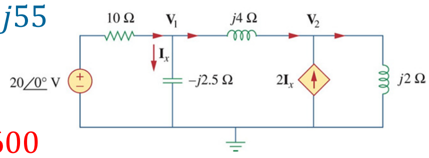
5

Example 10.1 cont

Determine i_x.

$$\begin{bmatrix} 2+j3 & j5 \\ 11 & 15 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 40 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 2+j3 & j5 \\ 11 & 15 \end{vmatrix} = (2+j3)\cdot 15 - j5\cdot 11$$
$$= (30 + j45) - j55$$
$$= 30 - j10$$



$$\Delta_1 = \begin{vmatrix} 40 & j5 \\ 0 & 15 \end{vmatrix} = 40\cdot 15 - 0 = 600$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{600}{31.62\angle -18.43°} = 18.97\angle 18.43°\ \text{V}$$

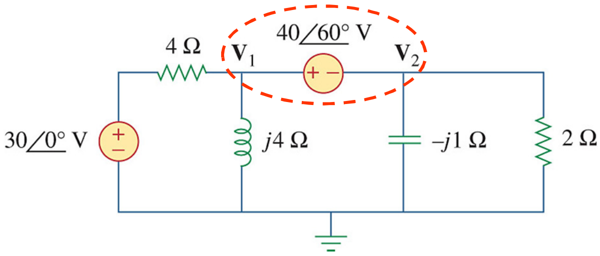
$$I_x = \frac{V_1}{-j2.5} = \frac{18.97\angle 18.43°}{2.5\angle -90°} = 7.59\angle 108.43°\ \text{A}$$

$$i_x = \underline{7.59\cos(4t + 108.43°)\ \text{A}}$$

6

PP 10.2 (Super Node)

Determine $\mathbf{V_1}$ and $\mathbf{V_2}$.



KCL at Super node:

$$\frac{V_1 - 30\angle 0°}{4} + \frac{V_1}{j4} + \frac{V_2}{-j1} + \frac{V_2}{2} = 0$$

$(\times 4)$ $\quad V_1 - 30 - jV_1 + j4V_2 + 2V_2 = 0$

$$(1-j)V_1 + (2+j4)V_2 = 30 \quad (1)$$

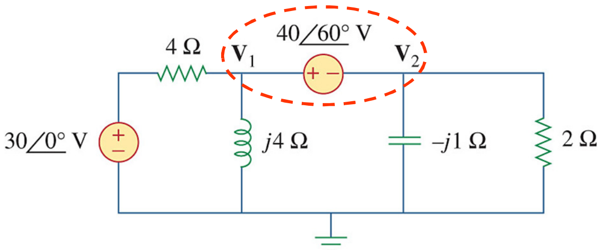
Super node: $\quad V_1 - V_2 = 40\angle 60°$

$$V_1 = V_2 + 40\angle 60° \quad (2)$$

7

PP 10.2 cont …

Determine $\mathbf{V_1}$ and $\mathbf{V_2}$.



(2) in (1)

$$(1-j)(V_2 + 40\angle 60°) + (2+j4)V_2 = 30$$

$$(1 - j + 2 + j4)V_2 + (1-j)(40\angle 60°) = 30$$

$$(3+j3)V_2 + \left(\sqrt{2}\angle -45°\right)(40\angle 60°) = 30$$

$$\left(\sqrt{18}\angle 45°\right)V_2 = 30 - \left(\sqrt{2}\cdot 40\angle 15°\right)$$

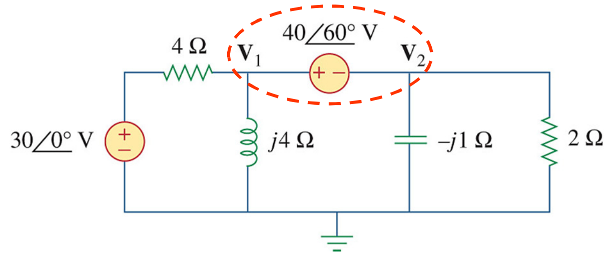
$$V_2 = \frac{30 - 54.64 - j14.64}{\sqrt{18}\angle 45°} = \frac{28.66\angle -149.28°}{4.24\angle 45°}$$

$$V_2 = 6.76\angle 165.72° \text{ V} \quad (3)$$

8

PP 10.2 cont …

Determine $\mathbf{V_1}$ and $\mathbf{V_2}$.



3 in 2

$$V_1 = V_2 + 40\angle 60°$$
$$= 6.76\angle 165.72° + 40\angle 60°$$
$$= -6.547 + j1.\dot{6} + 20 + j34.641$$
$$= 13.45 + j36.31$$
$$= 38.72\angle 69.67°\ \text{V}$$

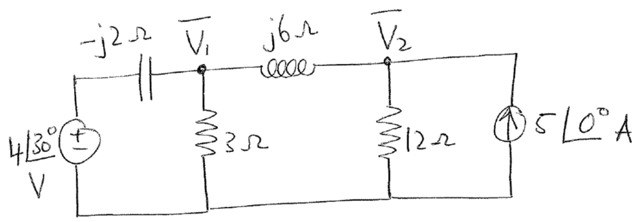
9

Example 1:

Set up the nodal analysis equations for $\mathbf{V_1}$ and $\mathbf{V_2}$. Write it in the following form:

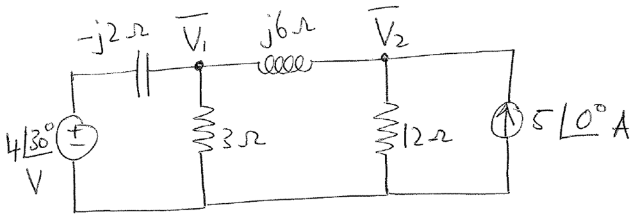
$(a_1 + jb_1)\mathbf{V_1} + (a_2 + jb_2)\mathbf{V_2} = A_3\angle\phi_3$

$(a_4 + jb_4)\mathbf{V_1} + (a_5 + jb_5)\mathbf{V_2} = A_6\angle\phi_6$



10

Exercise:



KCL at Node 1:

$$\frac{V_1 - 4\angle 30°}{-j2} + \frac{V_1}{3} + \frac{V_1 - V_2}{j6} = 0$$

$$(\times 6)\quad j3 \cdot (V_1 - 4\angle 30°) + 2V_1 + (-j)(V_1 - V_2) = 0$$

$$j3V_1 + 2V_1 - jV_1 + jV_2 = 12\angle 120°$$

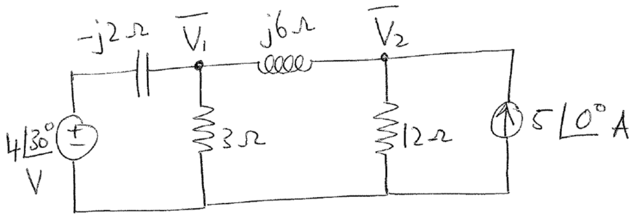
$$(2 - j2)V_1 + jV_2 = 12\angle 120°$$

$$(a_1 + jb_1)\mathbf{V_1} + (a_2 + jb_2)\mathbf{V_2} = A_3\angle\phi_3$$

$a_1 = 2$; $b_1 = -2$; $a_2 = 0$; $b_2 = 1$; $A_3 = 12$; $\phi_3 = 120°$

11

Exercise cont…



KCL at Node 2:

$$\frac{V_2 - V_1}{j6} + \frac{V_2}{12} - 5\angle 0° = 0$$

$$(\times 12)\quad (-j2)(V_2 - V_1) + V_2 - 60\angle 0° = 0$$

$$-j2V_2 + j2V_1 + V_2 = 60\angle 0°$$

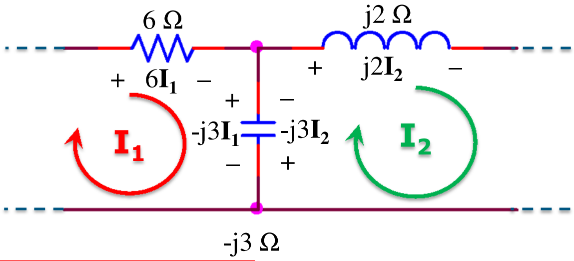
$$j2V_1 + (1 - 2j)V_2 = 60\angle 0°$$

$$(a_4 + jb_4)\mathbf{V_1} + (a_5 + jb_5)\mathbf{V_2} = A_6\angle\phi_6$$

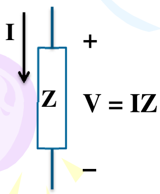
$a_4 = 0$; $b_4 = 2$; $a_5 = 1$; $b_5 = -2$; $A_6 = 60$; $\phi_6 = 0°$

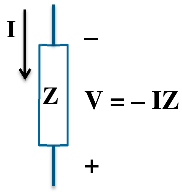
12

Ch. 10 L#2
10.3 Mesh Analysis



$\boxed{\ldots + 6\boldsymbol{I_1} + (-j3)(\boldsymbol{I_1} - \boldsymbol{I_2}) = 0}$ $\boxed{\ldots + j2\boldsymbol{I_2} + (-j3)(\boldsymbol{I_2} - \boldsymbol{I_1}) = 0}$

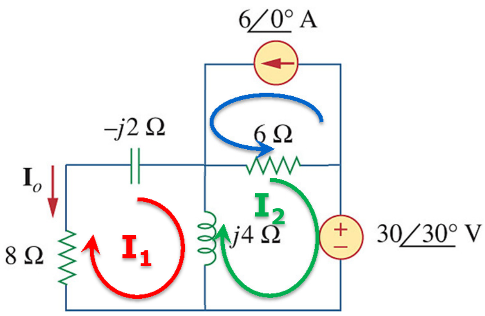




13

PP 10.3

Find $\mathbf{I}_o$.



KVL Loop 1:

$$8I_1 + (-j2)I_1 + j4(I_1 - I_2) = 0$$

$$(8 + j2)I_1 - j4I_2 = 0$$

$\boxed{\text{Note: } \mathbf{I_o} = -\mathbf{I_1}}$

$$I_2 = \frac{(8 + j2)I_1}{j4}$$

$$= (-j2 + 0.5)I_1 \quad (1)$$

KVL Loop 2:

$$j4(I_2 - I_1) + 6(I_2 + 6\angle 0°) + 30\angle 30° = 0$$

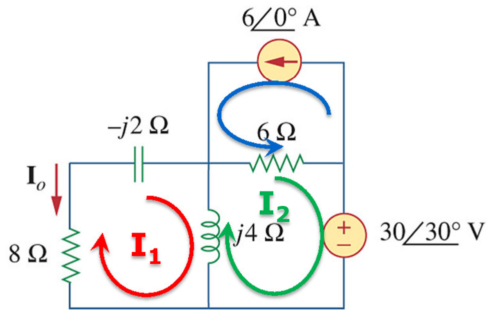
$$(-j4)I_1 + (6 + j4)I_2 = -36\angle 0° - 30\angle 30°$$

$$(-j4)I_1 + (6 + j4)I_2 = -62 - j15 = 63.77\square - 166.4\square \quad (2)$$

14

PP 10.3 cont…

Find $\mathbf{I}_o$.



1 in 2 :

$$(-j4)I_1 + (6+j4)(-j2+0.5)I_1 = 63.77\angle -166.4°$$

$$(-j4)I_1 + (-12j+3+8+j2)I_1 = 63.77\angle -166.4°$$

$$(11-j14)I_1 = 63.77\angle -166.4°$$

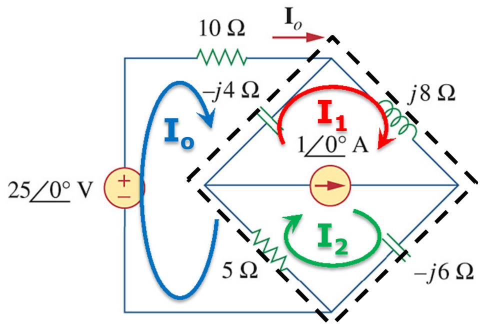
$$I_1 = \frac{63.77\angle -166.4°}{17\angle -51.84°} = 3.58\angle -114.55°$$

$$I_0 = -I_1 = \underline{3.58\angle 65.45°\ \text{A}}$$

15

PP 10.4

Find $\mathbf{I}_o$.



Super Mesh: $I_2 - I_1 = 1\angle 0°$

$$I_2 = 1\angle 0° + I_1 \quad (1)$$

Loop O:

$$-25\angle 0° + 10I_o + (-j4)(I_o - I_1) + 5(I_o - I_2) = 0$$

$$(15-j4)I_o + j4I_1 - 5I_2 = 25\angle 0°$$

$$(15-j4)I_o + j4I_1 - 5(1+I_1) = 25\angle 0°$$

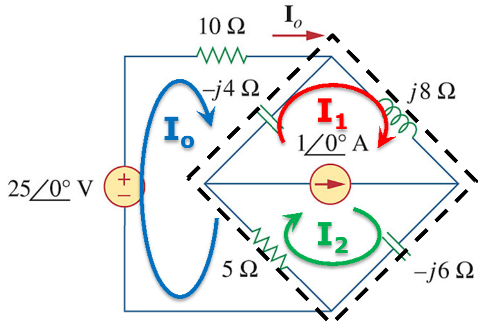
$$(15-j4)I_o + (-5+j4)I_1 = 30\angle 0°$$

$$I_1 = \frac{30\angle 0° - (15-j4)I_o}{(-5+j4)} \quad (2)$$

16

PP 10.4 cont…

Find $\mathbf{I_o}$.



Super Mesh: $I_2 - I_1 = 1\angle 0°$

$$I_2 = 1\angle 0° + I_1 \quad (1)$$

Super Mesh KVL:

$$5(I_2 - I_o) + (-j4)(I_1 - I_o) + j8I_1 - j6I_2 = 0$$

$$(-5 + j4)I_o + j4I_1 + (5 - j6)I_2 = 0$$

$$(-5 + j4)I_o + j4I_1 + (5 - j6)(1 + I_1) = 0$$

$$(-5 + j4)I_o + (5 - j2)I_1 = (-5 + j6) \quad (3)$$

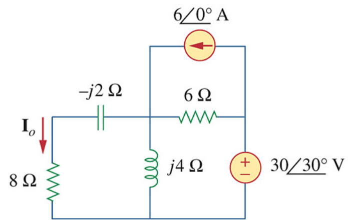
(2) in (3) **:** $(-5 + j4)I_o + (5 - j2)\left(\dfrac{30\angle 0° - (15 - j4)I_o}{(-5 + j4)}\right) = (-5 + j6)$

$$I_o = 2.54\angle 5.94° \text{ A}$$

17

PP 10.5 (Super Positioning)

Find $\mathbf{I_o}$.



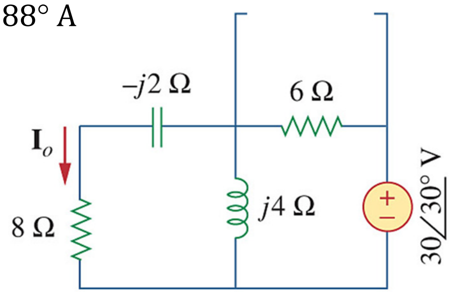
$\mathbf{I_o}$ due to 30∠30° V source:

$Z_{tot} = (8 - j2)//j4 + 6 = 8.636\ \angle 24.121°\ \Omega$

$$I_{tot} = \frac{30\angle 30°}{8.636\angle 24.121°} = 3.474\angle 5.88° \text{ A}$$

$$I_{o1} = 3.474\angle 5.88° \frac{j4}{8 - j2 + j4}$$

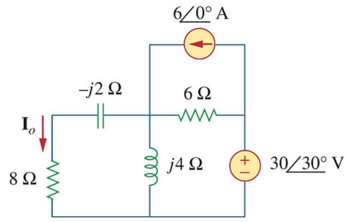
$$I_{o1} = 1.685\angle 81.843° \ A$$



18

PP 10.5 cont

Find $\mathbf{I}_o$.



$\mathbf{I}_o$ due to 6∠0° A source:

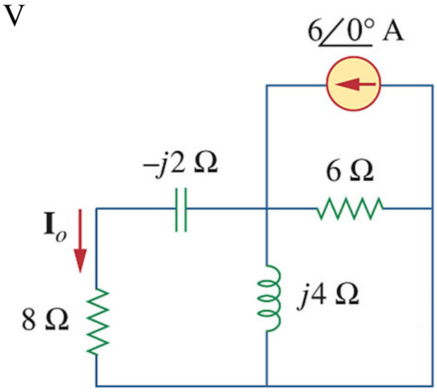
Z_{tot} = (8 – j2) // j4 // 6 = 2.779 ∠37.806° Ω

$V_{//}$ = 6∠0°×Z_{tot} = 16.674 ∠37.806° V

$$I_{o2} = \frac{16.674\angle 37.806°}{8 - j2}$$

$$I_{o2} = 2.022\angle 51.843°\ \text{A}$$

$$I_o = I_{o1} + I_{o2} = 3.58\angle 65.45°\ \text{A}$$

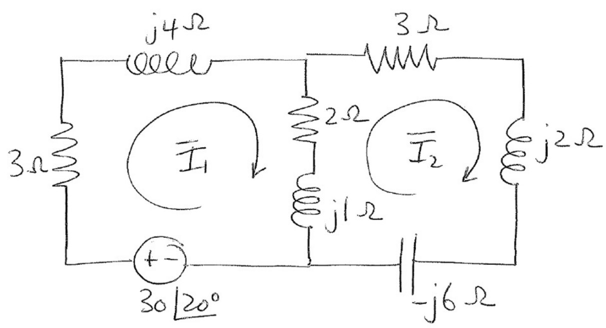


Example 2:

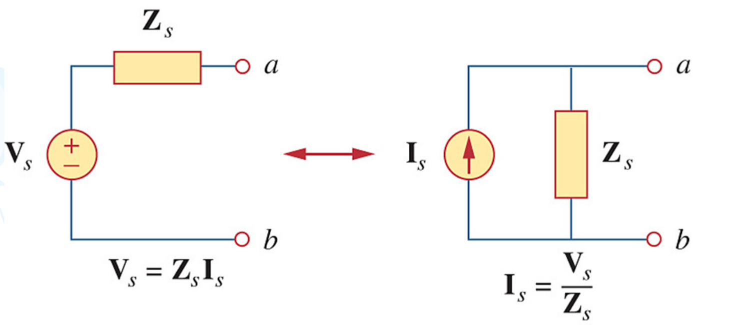
Set up the mesh equations for $\mathbf{I}_1$ and $\mathbf{I}_2$. Write it in the following form:

$(a_1 + jb_1)\mathbf{I}_1 + (a_2 + jb_2)\mathbf{I}_2 = A_3\angle\phi_3$

$(a_4 + jb_4)\mathbf{I}_1 + (a_5 + jb_5)\mathbf{I}_2 = A_6\angle\phi_6$



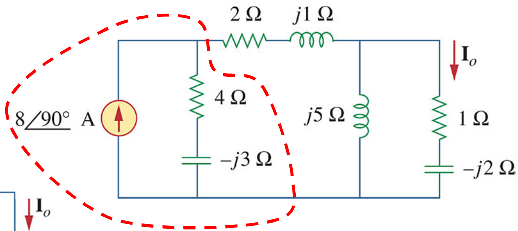
Ch. 10 L#3
10.5 Source Transformation
10.6 Thevenin & Norton

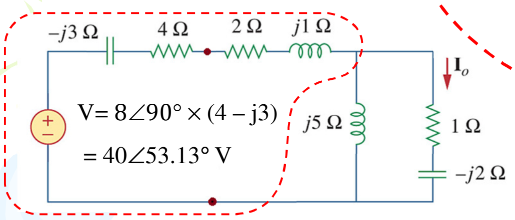


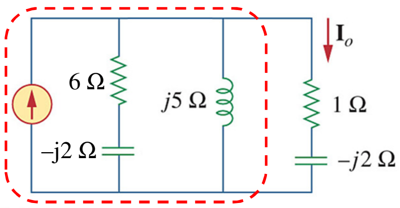
Thevenin Equivalent　　Norton Equivalent

PP 10.7

Find $\mathbf{I_o}$ using source transformation

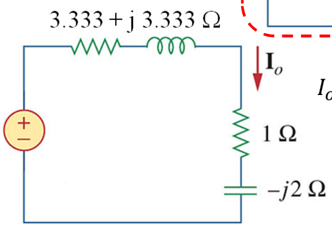






$$I = \frac{40\angle 53.13^\circ}{4+2+j1-j3} = 6.32\angle 71.57^\circ \text{ A}$$

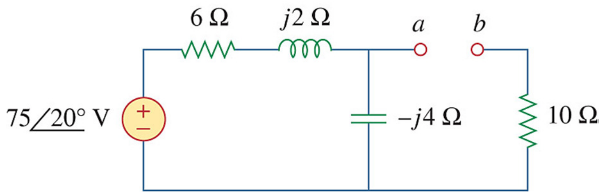
$$V = 6.32\angle 71.57^\circ \times (6-j2)//j5 = 29.81\angle 116.57^\circ \text{ V}$$



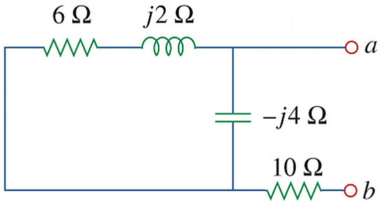
$$I_o = \frac{29.81\angle 116.57^\circ}{3.333+1+j(3.333-2)} = 6.58\angle 99.46^\circ \text{ A}$$

PP 10.8

Find the Thevenin equivalent at terminals a-b of the circuit



Thevenin Impedance: Z_{th}

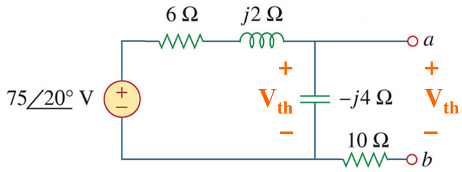


Z_{th} = (6+j2) // (-j4) + 10

= (12.4 – j3.2)

= 12.81∠-14.47° Ω

$V_{th} = V_{oc}$



Voltage Division:

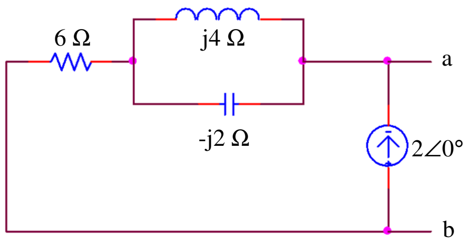
$$V_{th} = 75\angle 20° \frac{-j4}{6 + j2 - j4}$$

$= 47.43\angle -51.57°$ V

23

Example 3

Determine the Norton equivalent wrt terminals a-b



Thevenin Impedance: Z_{th}

I_{source} becomes open circuit

Z_{th} = j4 // (-j2) + 6

= (6 – j4)

= 7.21∠-33.7° Ω

$I_N = I_{sc} = 2\angle 0°$

24

Example 4

Solve the following Matrix $\begin{bmatrix} 3 & 2+j2 \\ 5+j5 & j2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5\angle 0° \end{bmatrix}$

$$V_1 = 1.01\angle -45° \text{ V}$$
$$V_2 = 1.07\angle 90° \text{ V}$$